E301 Energy at Home

E3.1.1 Energy-conversion efficiency

Efficiency is useful output energy divided by total input energy, or corresponding powers over the same interval. Ordinary conversion efficiency does not exceed 100%; the remaining energy goes into forms outside the intended useful output. Define what counts as useful and convert percentage efficiency to a fraction before substitution.

$$\eta_{\%} = \frac{E_{\text{useful}}}{E_{\text{input}}} \times 100\%$$

Percentage efficiency; useful output energy and total input energy in identical units. A fractional efficiency omits × 100%.

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E3.1.2 Useful heating power

For water of mass m warmed by ΔT in time t with no state change, useful heat is $Q = mc\Delta T$ and useful power is $P_{\text{useful}} = \frac{mc\Delta T}{t}$. With efficiency η, input power is $P_{\text{input}} = \frac{mc\Delta T}{\eta t}$, since efficiency is useful output divided by input. Convert minutes to seconds and use consistent mass and specific-heat units. With losses, water heating power is lower than appliance input power.

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E3.1.3 Interpreting energy labels

The textbook's energy-label example uses grades 1–5, with grade 1 most efficient and grade 5 least efficient. Energy use and operating time can also be used to calculate input power. Efficiency grade and total electricity consumption are different pieces of information. Compare within the product category and conditions provided in the question.

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E3.1.4 Hotplates, induction cookers and microwave ovens

A hotplate heats a resistive element and transfers energy to a pan. An induction cooker uses a changing magnetic field to induce currents and heating within a suitable metal pan. A microwave oven's oscillating electric field drives polar molecules in food, with interactions converting energy into internal energy. Their heating locations and mechanisms differ.

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E3.1.5 · E3.1.7 Energy flow in a heat pump

A heat pump uses external work to move energy from a colder region to a hotter one, opposite natural heat flow. Heat delivered to the hot side equals heat removed from the cold side plus work input. A refrigerator releases more heat at its back than it removes inside; it does not create energy.

$$Q_H = Q_C + W$$

Refrigerator/heat pump per cycle: Q_{H}: heat delivered to hot reservoir; Q_{C}: heat removed from cold reservoir; W: work input. All are positive energy magnitudes (J).

E3.1.6 The refrigeration cycle

In the evaporator, refrigerant absorbs indoor heat and vaporises. The compressor raises its pressure and temperature. The condenser releases heat outdoors and liquefies it. Pressure drops after the expansion valve, allowing partial evaporation and cooling before return to the evaporator. Match each component on the cycle diagram with its state changes and function.

E3.1.8 · E3.1.9 Cooling capacity and coefficient of performance

Cooling capacity is heat removed from the cold side per second, not electrical input power. Refrigerating coefficient of performance is heat removed divided by work input, or the corresponding power ratio. It can exceed 1 because the input moves heat rather than merely becoming the removed heat. Compare the same time interval.

$$P_{\text{cool}} = \frac{Q_C}{t}$$

$$\text{COP}_{\text{cool}} = \frac{P_{\text{cool}}}{P_{\text{in}}} = \frac{Q_C}{W}$$

Cooling capacity P_{cool} (W); Q_{C}: heat removed from the cold reservoir (J); t: s. Not the electrical input power.

Cooling coefficient of performance is dimensionless and may exceed 1. Q_{C}: heat removed; W: work input, both in J and over the same interval.

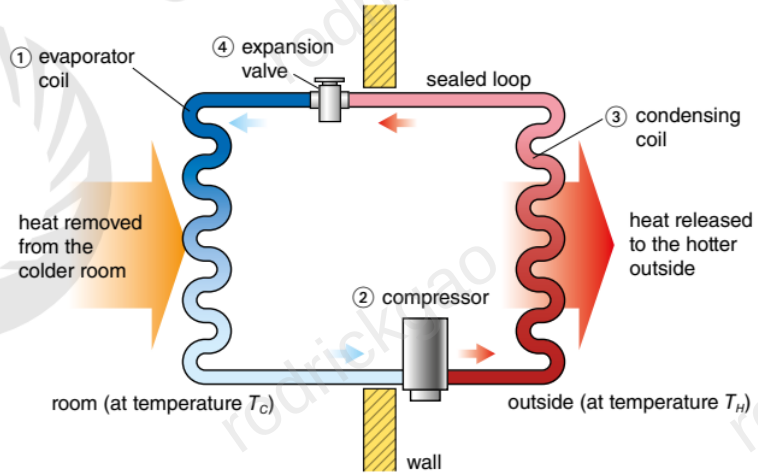


Fig 1.2d Cyclic process in an air conditioner

The evaporator absorbs heat, the compressor receives work, the condenser releases heat and the expansion valve lowers pressure.

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